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**Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: I Session: 2018-2019**

**Course Code**  
CCE-313

**Course Title**  
Computer Network

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**Credit: 03**

Time: 03 Hr

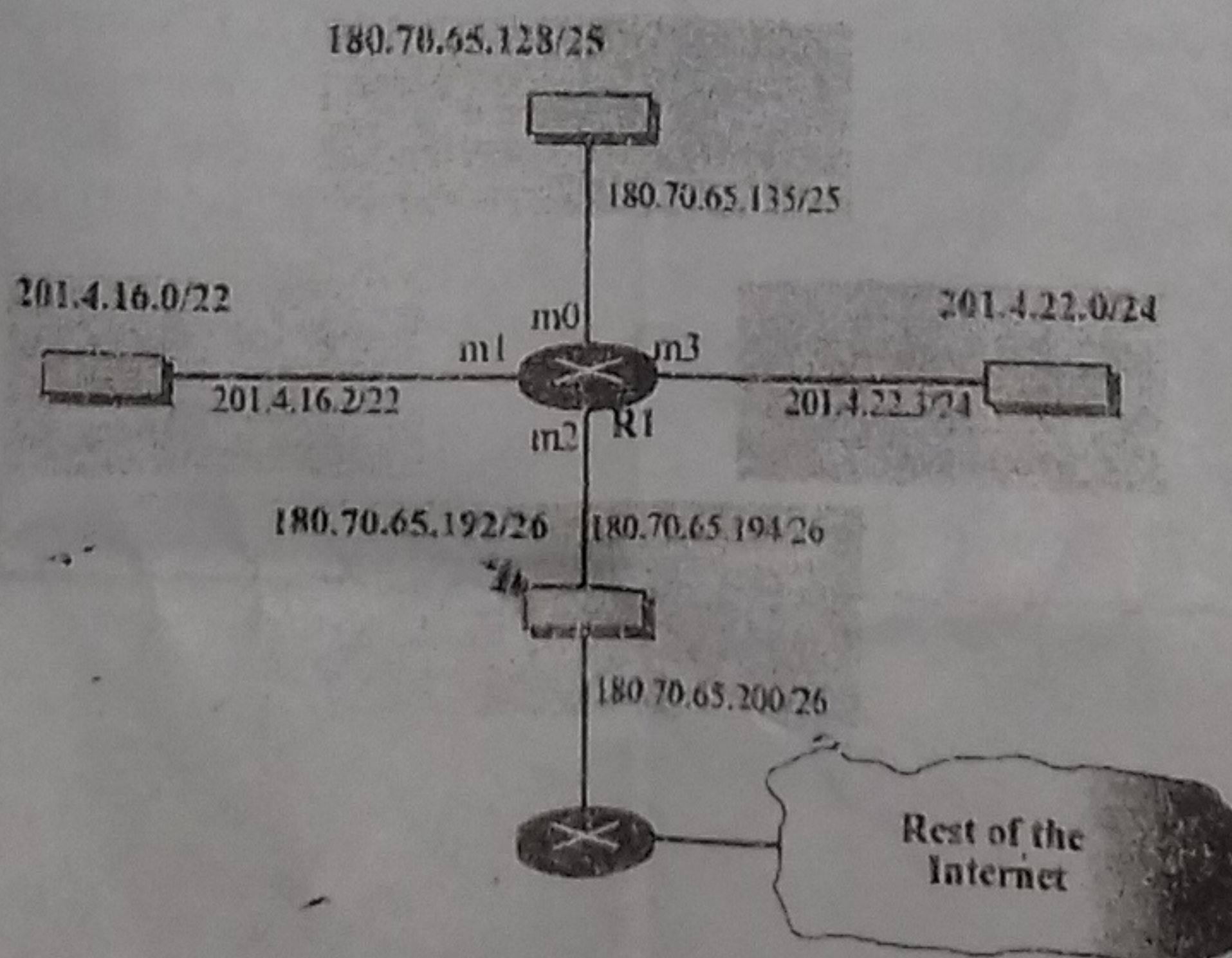
**Marks: 70**

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

- [A.] Describe the most popular wireless Internet access technologies today. Compare and contrast them.  
 b) List four access technologies. Classify each one as home access, enterprise access, or wide-area wireless access. DSL, ADSL, Ethernet, 4G, 5G  
 [B.] Write the duties/services and protocol of each specific OSI layer.  
 [C.] Consider an application that transmits data at a steady rate (for example, the sender generates an N-bit unit of data every k time units, where k is small and fixed). Also, when such an application starts, it will continue running for a relatively long period of time. Answer the following questions, briefly justifying your answer:  
 a. Would a packet-switched network or a circuit-switched network be more appropriate for this application? Why?  
 b. Suppose that a packet-switched network is used and the only traffic in this network comes from such applications as described above. Furthermore, assume that the sum of the application data rates is less than the capacities of each and every link.  
 Is some form of congestion control needed? Why?  
 [D.] Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates  $R_1 = 500 \text{ kbps}$ ,  $R_2 = 2 \text{ Mbps}$ , and  $R_3 = 1 \text{ Mbps}$ . 1000  
500k  
 a. Assuming no other traffic in the network, what is the throughput for the file transfer? —  
 b. Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?  
 c. Repeat (a) and (b), but now with  $R_2$  reduced to 100 kbps.

- |      |                                                                                                                                                                                                                                                                                                                                    |   |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| [A.] | How are the blocks allocated? What are the restrictions need to be applied to the allocated block.                                                                                                                                                                                                                                 | 3 |
| [B.] | One of the addresses in a block is 110.23.120.14/20. Find the number of addresses, the first address, and the last address in the block.                                                                                                                                                                                           | 3 |
| [C.] | An organization is granted a block of addresses with the beginning address 254.24.252.0/24. The organization needs to have 4 subblocks of addresses to use in its four subnets as shown below:<br>One subblock of 123 addresses.<br>One subblock of 62 addresses.<br>One subblock of 11 addresses.<br>One subblock of 5 addresses. | 6 |
| [D.] | Write the steps need to be carefully followed to guarantee the proper operation of the subnetworks.                                                                                                                                                                                                                                | 2 |

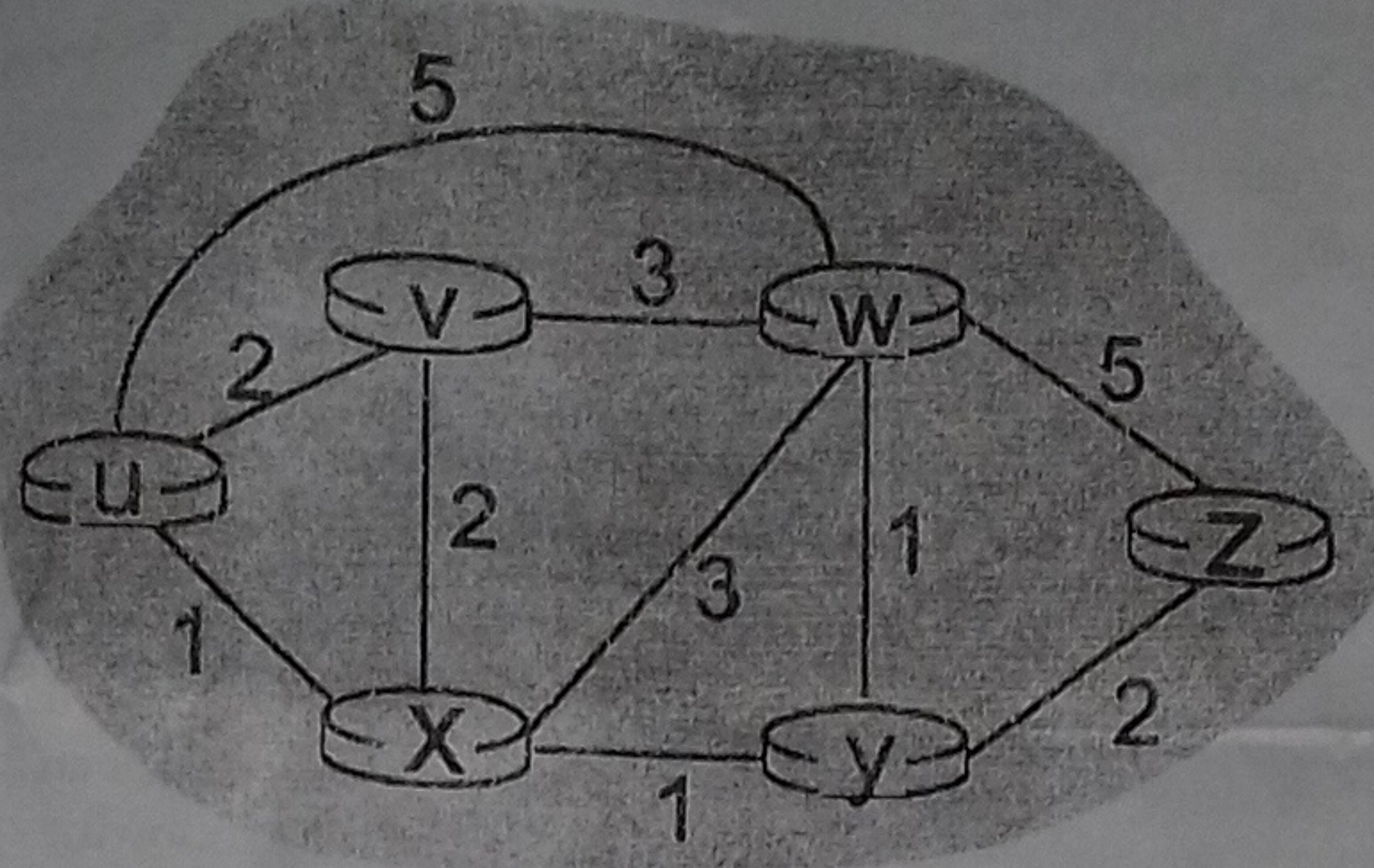
- [A.] Discuss the packet forwarding without Subnetting.  
[B.] Show the forwarding process if a packet arrives at R1 in below Figure with the destination address 180.70.65.140.



- [C.] Compare and contrast the properties of a centralized and a distributed routing algorithm. Give an example of a routing protocol that takes a centralized and a decentralized approach.



- [D.] Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from u to all network nodes. Show how the algorithm works by computing a table. 5



- 4 [A.] The following is a dump of a UDP header in hexadecimal format. 5  
0045DF000058FE20  
 a. What is the source port number?  
 b. What is the destination port number?  
 c. What is the total length of the user datagram?  
 d. What is the length of the data?  
 e. Is the packet directed from a client to a server or vice versa?  
 f. What is the client process?
- [B.] How do opening, closing, and shrinking of window occur in TCP? Give examples for both send and receive window. 4
- [C.] What are the services that a transport-layer protocol can offer to applications invoking it? Describe each with example. 3
- [D.] A sender sends a JPEG message. Show the MIME header. 2
- 5 [A.] Suppose, An FTP server has received a packet from an FTP client with IP address 153.2.7.9. The FTP server wants to verify that the FTP client is an authorized client. The FTP server can consult a file containing the list of authorized clients. However, the file consists only of domain names. The FTP server has only the IP address of the requesting client, which was the source IP address in the received IP datagram. The FTP server asks the resolver (DNS client) to send an inverse query to a DNS server to ask for the name of the FTP client. Show the query and response messages with values for each field. Discuss separately. 5
- [B.] Describe how Web caching can reduce the delay in receiving a requested object. Will Web caching reduce the delay for all objects requested by a user or for only some of the objects? Why? 4
- [C.] What are the services provided by the SSH protocol through port forwarding? Explain with example. 3
- [D.] Show the encoding for the INTEGER 1456 in case of SNMP. 2
- 6 [A.] Suppose a content provider, NetCinema, employs the third-party CDN company, KingCDN, to distribute its videos to its customers. On the NetCinema Web pages, each of its videos is assigned a URL that includes the string "video" and a unique identifier for the video itself; for example, *Transformers 7* might be assigned <http://video.netcinema.com/6Y7B23V>. If DASH is used, what will be the steps before the client can dynamically select chunks from the different versions of the video? 5
- [B.] Consider a new peer Alice that joins BitTorrent without possessing any chunks. Without any chunks, she cannot become a top-four uploader for any of the other peers, since she has nothing to upload. How then will Alice get her first chunk? 4
- [C.] Define Jitter. Explain the application of TFTP in conjunction with DHCP. 3
- [D.] Do you think H.323 is actually the same as SIP? Make a comparison between the two. 2